Chapter 5. Diagonalization

Eigenvalues, eigenvectors and invariant subspaces

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5.1 Eigenvalues and eigenvectors

Motivation. An operator is easiest to understand in a basis where its matrix is diagonal. Such a basis consists of eigenvectors, vectors that T only stretches, and the stretching factors are the zeros of the characteristic polynomial $\det(A - tI)$.

Definition (Diagonalizable)

A linear operator T on finite-dimensional V is **diagonalizable** if $[\mathsf{T}]_\beta$ is diagonal for some ordered basis β. A square matrix A is diagonalizable if L_A is, i.e. if A is similar to a diagonal matrix.

Definition (Eigenvector, eigenvalue)

A nonzero $v \in V$ is an **eigenvector** of T if $\mathsf{T}(v) = \lambda v$ for some scalar λ, the corresponding **eigenvalue**. For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, eigenvectors and eigenvalues are those of L_A: $Av = \lambda v$, $v \neq 0$.

Theorem 5.1

T is diagonalizable iff V has an ordered basis β of eigenvectors of T. If $\beta = \{v_1, \ldots, v_n\}$ is such a basis and $D = [\mathsf{T}]_\beta$, then D is diagonal and D_{jj} is the eigenvalue of v_j.

Proof. $[\mathsf{T}]_\beta$ is diagonal iff each column $[\mathsf{T}(v_j)]_\beta$ is $D_{jj}e_j$, i.e. $\mathsf{T}(v_j) = D_{jj}v_j$ for all j. Basis vectors are nonzero, so this says each v_j is an eigenvector with eigenvalue D_{jj}. ■

To diagonalize: find a basis of eigenvectors γ; then $Q^{-1}AQ$ is diagonal, where the columns of Q are the vectors of γ (Corollary to Theorem 2.23).

Example 1

$A = \left(\begin{smallmatrix} 1 & 3 \\ 4 & 2 \end{smallmatrix}\right)$: $A(1,-1)^t = -2(1,-1)^t$ and $A(3,4)^t = 5(3,4)^t$. With $Q = \left(\begin{smallmatrix} 1 & 3 \\ -1 & 4 \end{smallmatrix}\right)$, $Q^{-1}AQ = \left(\begin{smallmatrix} -2 & 0 \\ 0 & 5 \end{smallmatrix}\right)$.

Example 2

The rotation of $\mathbb{R}^2$ by $\pi/2$ has no eigenvectors (no nonzero vector is mapped to a multiple of itself), so it is not diagonalizable.

Example 3

On $C^\infty(\mathbb{R})$, $\mathsf{T}(f) = f'$ has $\mathsf{T}(f) = \lambda f$ iff $f = ce^{\lambda t}$ (as in Theorem 2.30): every real λ is an eigenvalue, with eigenvectors $ce^{\lambda t}$, $c \neq 0$.

Theorem 5.2

λ is an eigenvalue of $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ iff $\det(A - \lambda I_n) = 0$.

Proof. λ is an eigenvalue iff $(A - \lambda I)v = 0$ for some $v \neq 0$, iff $A - \lambda I$ is not invertible, iff $\det(A - \lambda I) = 0$ (Corollary to Theorem 4.7). ■

Definition (Characteristic polynomial)

$f(t) = \det(A - tI_n)$ is the **characteristic polynomial** of A. For an operator T on n-dimensional V, it is that of $[\mathsf{T}]_\beta$, independent of β since similar matrices have the same characteristic polynomial: $\det(Q^{-1}AQ - tI) = \det(Q^{-1}(A - tI)Q) = \det(A - tI)$. Write $f(t) = \det(\mathsf{T} - t\mathsf{I})$.

Example 4

$A = \left(\begin{smallmatrix} 1 & 1 \\ 4 & 1 \end{smallmatrix}\right)$: $f(t) = (1-t)^2 - 4 = (t-3)(t+1)$, so the eigenvalues are 3 and -1.

Example 5

$\mathsf{T}(f) = f + (x+1)f'$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{smallmatrix}\right)$, so $f(t) = -(t-1)(t-2)(t-3)$ and the eigenvalues are $1, 2, 3$.

Theorem 5.3

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$: (a) $f(t)$ has degree n and leading coefficient $(-1)^n$;
(b) A has at most n distinct eigenvalues.

Proof [Supplement].

(a) Induction on n, expanding $\det(A - tI)$ along the first row: the term $(A_{11} - t)\det((\widetilde{A - tI})_{11})$ has degree n and leading coefficient $(-1) \cdot (-1)^{n-1}$ by induction, while the other terms contain a minor missing two diagonal entries, of degree $\leq n - 2$.

(b) A nonzero polynomial of degree n has at most n zeros (Section 1.6.1). ■

Theorem 5.4

If λ is an eigenvalue of T, then v is an eigenvector for λ iff $v \neq 0$ and $v \in \mathrm{N}(\mathsf{T} - \lambda\mathsf{I})$.

Proof. $\mathsf{T}(v) = \lambda v$ iff $(\mathsf{T} - \lambda\mathsf{I})(v) = 0$. ■

Example 6

For A of Example 4: $\mathrm{N}(A - 3I) = \mathrm{span}\{(1,2)^t\}$ and $\mathrm{N}(A + I) = \mathrm{span}\{(1,-2)^t\}$. So $\beta = \{(1,2),(1,-2)\}$ is a basis of eigenvectors and $Q^{-1}AQ = \mathrm{diag}(3,-1)$ for $Q = \left(\begin{smallmatrix} 1 & 1 \\ 2 & -2 \end{smallmatrix}\right)$.

Example 7

For T of Example 5, solving $(A - \lambda I)x = 0$ gives eigenvectors 1, $1 + x$, $1 + 2x + x^2 = (1+x)^2$ for $\lambda = 1, 2, 3$. In the basis $\{1, 1+x, (1+x)^2\}$, $[\mathsf{T}] = \mathrm{diag}(1,2,3)$.

In general one computes eigenvectors of T through coordinates: v is an eigenvector of T iff $[v]_\beta$ is one of $[\mathsf{T}]_\beta$.

5.2 Diagonalizability

Motivation. Eigenvectors for distinct eigenvalues are independent, so n distinct eigenvalues guarantee a diagonal form. In general, diagonalizability means that the characteristic polynomial splits

and that each eigenspace is as large as the multiplicity allows, i.e. V is the direct sum of the eigenspaces.

Theorem 5.5

Eigenvectors $v_1, \dots, v_k$ of T for distinct eigenvalues $\lambda_1, \dots, \lambda_k$ are linearly independent.

Proof. Induction on k; $k = 1$ holds as $v_1 \neq 0$. If $\sum_{i=1}^{k} a_i v_i = 0$, apply $\mathsf{T} - \lambda_k \mathsf{I}$: $\sum_{i<k} a_i(\lambda_i - \lambda_k)v_i = 0$. By induction $a_i(\lambda_i - \lambda_k) = 0$, so $a_i = 0$ for $i < k$; then $a_k v_k = 0$ gives $a_k = 0$. ■

Corollary

If $\dim V = n$ and T has n distinct eigenvalues, T is diagonalizable.

Proof. One eigenvector per eigenvalue gives n independent vectors (Theorem 5.5), a basis (Corollary 2 of Theorem 1.10); apply Theorem 5.1. ■

Example 1

$A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has eigenvalues $0, 2$, hence is diagonalizable. The converse of the Corollary fails: I is diagonal with one eigenvalue.

Definition (Splitting)

$f \in \mathsf{P}(\mathbb{F})$ **splits over** $\mathbb{F}$ if $f(t) = c(t - a_1) \cdots (t - a_n)$ with $c, a_i \in \mathbb{F}$. E.g. $(t^2 + 1)(t - 2)$ splits over $\mathbb{C}$ but not over $\mathbb{R}$.

Theorem 5.6

The characteristic polynomial of a diagonalizable operator splits.

Proof. If $[\mathsf{T}]_\beta = D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, then $f(t) = \det(D - tI) = (\lambda_1 - t) \cdots (\lambda_n - t) = (-1)^n (t - \lambda_1) \cdots (t - \lambda_n)$. ■

Definition (Multiplicity; eigenspace)

The (algebraic) **multiplicity** of an eigenvalue λ is the largest k with $(t - \lambda)^k \mid f(t)$. The **eigenspace** is $E_\lambda = \{x : \mathsf{T}(x) = \lambda x\} = \mathrm{N}(\mathsf{T} - \lambda \mathsf{I})$.

Example 2

$A = \left(\begin{smallmatrix} 3 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 4 \end{smallmatrix}\right)$ has $f(t) = -(t - 3)^2(t - 4)$: $\lambda = 3$ has multiplicity 2, $\lambda = 4$ multiplicity 1.

Theorem 5.7

If λ is an eigenvalue of multiplicity m of T on finite-dimensional V, then $1 \leq \dim E_\lambda \leq m$.

Proof. Let $\{v_1, \dots, v_p\}$ be a basis of E_λ ($p \geq 1$), extended to an ordered basis β of V. Then $A = [\mathsf{T}]_\beta = \left(\begin{smallmatrix} \lambda I_p & B \\ O & C \end{smallmatrix}\right)$, so by fact 8 of Section 4.4, $f(t) = \det(A - tI) = (\lambda - t)^p \det(C - tI)$. Hence $(t - \lambda)^p \mid f$, and $p \leq m$. ■

Example 3

$\mathsf{T}(f) = f'$ on $\mathsf{P}_2(\mathbb{R})$: $[\mathsf{T}] = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, $f(t) = -t^3$, and E_0 is the constants, of dimension $1 < 3$. T is not diagonalizable.

Example 4

$\mathsf{T}(a_1, a_2, a_3) = (4a_1 + a_3,\ 2a_1 + 3a_2 + 2a_3,\ a_1 + 4a_3)$ has $f(t) = -(t-5)(t-3)^2$, $E_5 = \text{span}\{(1,2,1)\}$ and $E_3 = \text{span}\{(0,1,0), (-1,0,1)\}$. The union of these bases is a basis of eigenvectors.

Lemma

Let $\lambda_1, \ldots, \lambda_k$ be distinct eigenvalues of T and $v_i \in E_{\lambda_i}$. If $v_1 + \cdots + v_k = 0$, then all $v_i = 0$.

Proof. Otherwise the nonzero v_i are eigenvectors for distinct eigenvalues with a nontrivial relation (all coefficients 1), contradicting Theorem 5.5. ■

Theorem 5.8

If $\lambda_1, \ldots, \lambda_k$ are distinct eigenvalues of T and each $S_i \subseteq E_{\lambda_i}$ is finite and linearly independent, then $S = S_1 \cup \cdots \cup S_k$ is linearly independent.

Proof. Let $S_i = \{v_{i1}, \ldots, v_{in_i}\}$ and $\sum_{i,j} a_{ij} v_{ij} = 0$. Put $w_i = \sum_j a_{ij} v_{ij} \in E_{\lambda_i}$. Then $\sum w_i = 0$, so each $w_i = 0$ by the Lemma, and independence of S_i gives $a_{ij} = 0$. ■

Theorem 5.9

Let T on finite-dimensional V have splitting characteristic polynomial and distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ with multiplicities m_i.
(a) T is diagonalizable iff $m_i = \dim E_{\lambda_i}$ for all i.
(b) If T is diagonalizable and β_i is an ordered basis of E_{λ_i}, then $\beta = \beta_1 \cup \cdots \cup \beta_k$ is an ordered basis of V consisting of eigenvectors.

Proof. Let $n = \dim V$, $d_i = \dim E_{\lambda_i}$. As f splits, $\sum m_i = n$.

Step 1 (only if). Let β be a basis of eigenvectors and $\beta_i' = \beta \cap E_{\lambda_i}$, with n_i vectors. Then $n_i \le d_i \le m_i$, and $\sum n_i = n$ since every vector of β is in some E_{λ_i}. So $n = \sum n_i \le \sum d_i \le \sum m_i = n$, forcing $d_i = m_i$.

Step 2 (if) and (b). If $d_i = m_i$, then β in (b) is independent by Theorem 5.8 with $\sum d_i = \sum m_i = n$ vectors, hence a basis of eigenvectors. Under the hypothesis of (b), Step 1 shows $d_i = m_i$, and the same argument applies. ■

Test for diagonalization. T is diagonalizable iff (1) its characteristic polynomial splits and (2) for each eigenvalue, $m_i = \dim E_{\lambda_i} = n - \text{rank}(\mathsf{T} - \lambda_i \mathsf{I})$.

Example 5

$A = \left(\begin{smallmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{smallmatrix}\right)$: $\lambda = 3$ has multiplicity 2 but $\text{rank}(A - 3I) = 2$, so $\dim E_3 = 1$; A is not diagonalizable.

Example 6

$\mathsf{T}(f) = f(1) + f'(0)x + (f'(0) + f''(0))x^2$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $[\mathsf{T}] = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{smallmatrix}\right)$, $f(t) = -(t-1)^2(t-2)$, $E_1 = \operatorname{span}\{1, -x + x^2\}$ (dimension 2), $E_2 = \operatorname{span}\{1 + x^2\}$. So T is diagonalizable with basis $\{1, -x + x^2, 1 + x^2\}$.

Example 7

$A = \left(\begin{smallmatrix} 0 & -2 \\ 1 & 3 \end{smallmatrix}\right)$ has eigenvalues $1, 2$ with eigenvectors $(-2, 1)^t, (-1, 1)^t$. With $Q = \left(\begin{smallmatrix} -2 & -1 \\ 1 & 1 \end{smallmatrix}\right)$, $A = QDQ^{-1}$, $D = \operatorname{diag}(1, 2)$, so

$$A^n = QD^nQ^{-1} = \begin{pmatrix} 2 - 2^n & 2 - 2^{n+1} \\ -1 + 2^n & -1 + 2^{n+1} \end{pmatrix}.$$

5.2.1 Systems of differential equations

Consider $x' = Ax$ with $A = \left(\begin{smallmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ -1 & -1 & 1 \end{smallmatrix}\right)$, $x = (x_1, x_2, x_3)^t$ real functions. A has eigenvalues 2 (eigenvectors $(-1, 1, 0)$, $(-1, 0, 1)$) and 4 (eigenvector $(-1, -2, 1)$), so $Q^{-1}AQ = D = \operatorname{diag}(2, 2, 4)$ with Q having these columns. For $y = Q^{-1}x$, $y' = Q^{-1}x' = Dy$, i.e. $y_1' = 2y_1$, $y_2' = 2y_2$, $y_3' = 4y_3$, so $y_i = c_i e^{\lambda_i t}$. Hence

$$x = Qy = e^{2t}\left(c_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right) + c_3 e^{4t} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}.$$

5.2.2 Direct sums

Definition (Sum of subspaces)

$W_1 + \cdots + W_k = \sum_i W_i = \{v_1 + \cdots + v_k : v_i \in W_i\}$, a subspace.

Example 8

In $\mathbb{R}^3$, the xy-plane W_1 and the yz-plane W_2 satisfy $\mathbb{R}^3 = W_1 + W_2$: $(a, b, c) = (a, 0, 0) + (0, b, c)$. The representation is not unique.

Definition (Direct sum)

$V = W_1 \oplus \cdots \oplus W_k$ if $V = \sum_i W_i$ and $W_j \cap \sum_{i \neq j} W_i = \{0\}$ for each j.

Example 9

$\mathbb{R}^4 = W_1 \oplus W_2 \oplus W_3$ for $W_1 = \{(a, b, 0, 0)\}$, $W_2 = \{(0, 0, c, 0)\}$, $W_3 = \{(0, 0, 0, d)\}$.

Theorem 5.10

For subspaces $W_1, \ldots, W_k$ of finite-dimensional V, the following are equivalent:
(a) $V = W_1 \oplus \cdots \oplus W_k$;

(b) $V = \sum W_i$, and $v_1 + \cdots + v_k = 0$ with $v_i \in W_i$ implies all $v_i = 0$;
(c) each $v \in V$ is uniquely $v = v_1 + \cdots + v_k$ with $v_i \in W_i$;
(d) if γ_i is an ordered basis of W_i for each i, then $\gamma_1 \cup \cdots \cup \gamma_k$ is an ordered basis of V;
(e) for each i there is an ordered basis γ_i of W_i with $\gamma_1 \cup \cdots \cup \gamma_k$ an ordered basis of V.

Proof.

Step (a)⇒(b). If $\sum v_i = 0$, then $-v_j = \sum_{i \neq j} v_i \in W_j \cap \sum_{i \neq j} W_i = \{0\}$.

Step (b)⇒(c). If $v = \sum v_i = \sum w_i$, then $\sum (v_i - w_i) = 0$, so $v_i = w_i$.

Step (c)⇒(d). The union generates V since $V = \sum W_i$. If a linear combination of all the basis vectors is 0, group it as $\sum_i u_i = 0$ with $u_i \in W_i$ a combination of γ_i; uniqueness for $0 = \sum 0$ gives $u_i = 0$, and independence of γ_i gives zero coefficients.

Step (d)⇒(e). Bases exist.

Step (e)⇒(a). $V = \operatorname{span}(\bigcup \gamma_i) \subseteq \sum W_i$. If $v \in W_j \cap \sum_{i \neq j} W_i$, write v as a combination of γ_j and also of $\bigcup_{i \neq j} \gamma_i$; subtracting gives a relation among the vectors of the basis $\bigcup \gamma_i$, so all coefficients vanish and $v = 0$. ■

Theorem 5.11

T on finite-dimensional V is diagonalizable iff V is the direct sum of the eigenspaces of T.

Proof. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues. If T is diagonalizable, with γ_i a basis of E_{λ_i}, then $\bigcup \gamma_i$ is a basis of V (Theorem 5.9(b)), so $V = \bigoplus E_{\lambda_i}$ by Theorem 5.10(e)⇒(a). Conversely, if $V = \bigoplus E_{\lambda_i}$, then by (a)⇒(d) the union of bases of the E_{λ_i} is a basis of eigenvectors. ■

Example 10

$\mathsf{T}(a, b, c, d) = (a, b, 2c, 3d)$ on $\mathbb{R}^4$: $E_1 = W_1$, $E_2 = W_2$, $E_3 = W_3$ of Example 9, and $\mathbb{R}^4 = E_1 \oplus E_2 \oplus E_3$.

5.3 Matrix limits and Markov chains *

Motivation. Powers A^m describe repeated transitions, as in population movements or Markov chains. Whether A^m converges is decided by the eigenvalues. For a regular transition matrix the limit exists and every column of it is the unique stationary probability vector.

Definition (Matrix limit)

$A_m \to L$ ($n \times p$, complex entries) if $(A_m)_{ij} \to L_{ij}$ for all i, j; write $\lim_m A_m = L$.

Example 1

$$A_m = \begin{pmatrix} 1 - \frac{1}{m} & (\frac{3}{4})^m \\ \frac{3m^2}{m^2+1} + i\frac{2m+1}{m-1} & (1+\frac{1}{m})^m \end{pmatrix} \to \begin{pmatrix} 1 & 0 \\ 3+2i & e \end{pmatrix}.$$

Theorem 5.12

If $A_m \to L$ ($n \times p$), then $PA_m \to PL$ and $A_m Q \to LQ$ for any $P \in \mathrm{M}_{r \times n}(\mathbb{C})$, $Q \in \mathrm{M}_{p \times s}(\mathbb{C})$.

Proof. $(PA_m)_{ij} = \sum_k P_{ik}(A_m)_{kj} \to \sum_k P_{ik}L_{kj}$, a finite sum of convergent sequences; similarly for A_mQ. ∎

Corollary

If $A^m \to L$ and Q is invertible, then $(QAQ^{-1})^m \to QLQ^{-1}$.

Proof. $(QAQ^{-1})^m = QA^mQ^{-1}$; apply Theorem 5.12 twice. ∎

Let $S = \{\lambda \in \mathbb{C} : |\lambda| < 1\} \cup \{1\}$. For $\lambda \in \mathbb{C}$, λ^m converges iff $\lambda \in S$.

Theorem 5.13

For $A \in \mathrm{M}_{n\times n}(\mathbb{C})$, $\lim A^m$ exists iff (a) every eigenvalue of A lies in S, and (b) if 1 is an eigenvalue, $\dim E_1$ equals the multiplicity of 1.

Proof [Supplement] (the book gives part of it and cites the rest; we use Theorems 5.23 and 5.24 of Section 5.4).

Step 1 (generalized eigenspaces). Over $\mathbb{C}$, $f(t) = \pm\prod_i (t-\lambda_i)^{m_i}$. Let $K_i = \mathrm{N}((A-\lambda_i I)^{m_i})$; it is L_A-invariant. The polynomials $g_i = \prod_{j\neq i}(t-\lambda_j)^{m_j}$ have no common zero, so $\sum_i p_i g_i = 1$ for some polynomials p_i (Appendix E). Put $P_i = p_i(A)g_i(A)$. Then $\sum P_i = I$, and $(A-\lambda_i I)^{m_i}P_i = \pm p_i(A)f(A) = O$ (Cayley–Hamilton), so $\mathrm{R}\,P_i \subseteq K_i$ and $\mathbb{C}^n = \sum K_i$. If $\sum v_i = 0$ with $v_i \in K_i$, apply $p_j(A)g_j(A)$: it kills v_i for $i \neq j$ (because $(t-\lambda_i)^{m_i} \mid g_j$), so $P_j v_j = 0$; since $v_j = \sum_i P_i v_j = P_j v_j$, $v_j = 0$. So $\mathbb{C}^n = \bigoplus K_i$ (Theorem 5.10).

Step 2 ($\dim K_i = m_i$). On K_i, $A - \lambda_i I$ is nilpotent, so λ_i is the only eigenvalue of $A|_{K_i}$. By Theorem 5.24, f is the product of the characteristic polynomials of the $A|_{K_i}$, so $\dim K_i = m_i$.

Step 3 (sufficiency). On K_i write $A = \lambda_i I + N$ with $N^{m_i} = O$. Then $A^m = \sum_{k<m_i}\binom{m}{k}\lambda_i^{m-k}N^k$. If $|\lambda_i| < 1$, then $\binom{m}{k}|\lambda_i|^{m-k} \to 0$ for each fixed k, so $A^m \to O$ on K_i. If $\lambda_i = 1$, (b) and $E_1 \subseteq K_1$, $\dim K_1 = m_1 = \dim E_1$ give $K_1 = E_1$, where $A = I$. Choosing a basis adapted to $\bigoplus K_i$, A is similar to a block-diagonal matrix whose powers converge; apply the Corollary to Theorem 5.12.

Step 4 (necessity of (a)). If $Av = \lambda v$, $v \neq 0$, and $L = \lim A^m$, then $Lv = \lim A^m v = \lim \lambda^m v$ (Theorem 5.12), so λ^m converges and $\lambda \in S$.

Step 5 (necessity of (b)). If $\dim E_1 < m_1 = \dim K_1$, then $A - I$ is nilpotent but nonzero on K_1, so some $w \in K_1$ has $u = (A-I)w \neq 0$ and $(A-I)u = 0$. Then $A^m w = w + mu$ diverges. ∎

Theorem 5.14

If every eigenvalue of $A \in \mathrm{M}_{n\times n}(\mathbb{C})$ lies in S and A is diagonalizable, then $\lim A^m$ exists.

Proof. $A = QDQ^{-1}$ with $D = \mathrm{diag}(\lambda_i)$, $\lambda_i \in S$, so $D^m = \mathrm{diag}(\lambda_i^m)$ converges, and $A^m = QD^mQ^{-1}$ converges by the Corollary to Theorem 5.12. ∎

Markov chains. A **transition matrix** (stochastic matrix) has nonnegative real entries and columns summing to 1: A_{ij} is the probability of moving from state j to state i in one step. A **probability vector** has nonnegative coordinates summing to 1. If P gives the initial distribution, A^mP gives the distribution after m steps, and $(A^m)_{ij}$ is the probability of going from j to i in m steps. Let $u = (1,\dots,1)^t$.

Theorem 5.15

Let M be $n \times n$ with nonnegative real entries and $v \in \mathbb{R}^n$ with nonnegative coordinates. Then (a) M is a transition matrix iff $M^t u = u$;

(b) v is a probability vector iff $u^t v = (1)$.

Proof. $(M^t u)_j$ is the j-th column sum of M, and $u^t v$ is the coordinate sum of v. ■

Corollary

(a) Products of transition matrices are transition matrices; in particular so are their powers.
(b) A transition matrix times a probability vector is a probability vector.

Proof. Nonnegativity is preserved. (a) $(MN)^t u = N^t M^t u = N^t u = u$.
(b) $u^t(Mv) = (M^t u)^t v = u^t v = 1$. ■

Definition (Regular)

A transition matrix is **regular** if some power of it has only positive entries.

Example 2

$\left(\begin{smallmatrix} 0.90 & 0.02 \\ 0.10 & 0.98 \end{smallmatrix}\right)$ (city/suburb migration) is regular. A matrix with an absorbing state, i.e. a column e_1, is not, since the first column of every power is e_1.

Definition (Gerschgorin disks)

For $A \in \mathrm{M}_{n\times n}(\mathbb{C})$, $\rho_i(A) = \sum_j |A_{ij}|$ (row sums), $\nu_j(A) = \sum_i |A_{ij}|$ (column sums), $\rho(A) = \max_i \rho_i(A)$, $\nu(A) = \max_j \nu_j(A)$. The i-th **Gerschgorin disk** C_i has center A_{ii} and radius $r_i = \rho_i(A) - |A_{ii}|$.

Example 3

For $A = \left(\begin{smallmatrix} 1 & 2 & -3 \\ 0 & 4 & 1 \\ 2 & -1 & 5 \end{smallmatrix}\right)$: $\rho_i = 6, 5, 8$ and $\nu_j = 3, 7, 9$, so $\rho(A) = 8$, $\nu(A) = 9$.

Theorem 5.16 (Gerschgorin's disk theorem)

Every eigenvalue of $A \in \mathrm{M}_{n\times n}(\mathbb{C})$ lies in some Gerschgorin disk.

Proof. Let $Av = \lambda v$, $v \neq 0$, and choose k with $|v_k|$ maximal. From $\lambda v_k = \sum_j A_{kj} v_j$, $|\lambda - A_{kk}||v_k| = |\sum_{j\neq k} A_{kj} v_j| \leq \sum_{j\neq k} |A_{kj}||v_k| = r_k |v_k|$, so $|\lambda - A_{kk}| \leq r_k$. ■

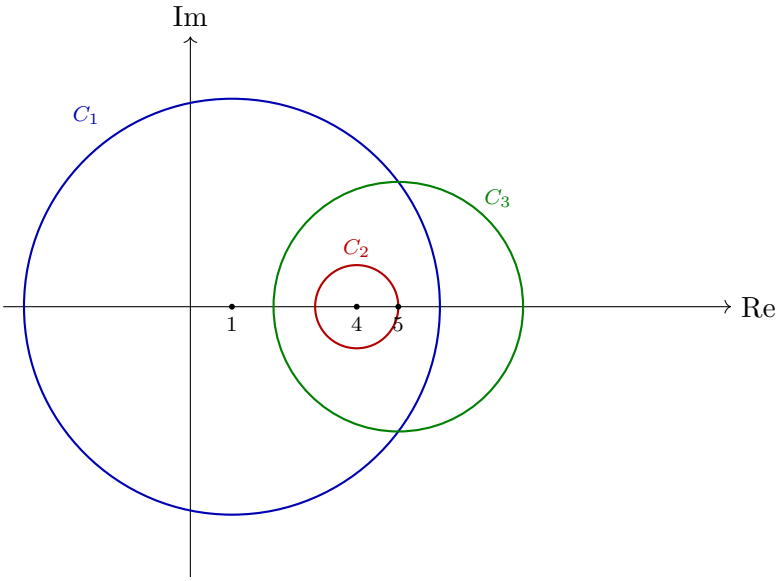


Figure. *Gerschgorin disks of the matrix of Example 3: centers* $1, 4, 5$*, radii* $5, 1, 3$*. All eigenvalues lie in their union.*

Corollary 1

$|\lambda| \le \rho(A)$ for every eigenvalue λ of A.

Proof. $|\lambda| \le |\lambda - A_{kk}| + |A_{kk}| \le r_k + |A_{kk}| = \rho_k(A) \le \rho(A)$. ■

Corollary 2

$|\lambda| \le \min\{\rho(A), \nu(A)\}$.

Proof. λ is an eigenvalue of A^t (same characteristic polynomial, Theorem 4.8), and $\rho(A^t) = \nu(A)$; apply Corollary 1 to both. ■

Corollary 3

Every eigenvalue λ of a transition matrix satisfies $|\lambda| \le 1$.

Proof. $\nu(A) = 1$; apply Corollary 2. ■

Theorem 5.17

Every transition matrix has 1 as an eigenvalue.

Proof. $A^t u = u$ (Theorem 5.15), so 1 is an eigenvalue of A^t, hence of A. ■

Theorem 5.18

Let $A \in \mathrm{M}_{n\times n}(\mathbb{C})$ have only positive entries, and let λ be an eigenvalue with $|\lambda| = \rho(A)$. Then $\lambda = \rho(A)$ and $\{u\}$ is a basis of E_λ.

Proof. Let $Av = \lambda v$, $v \neq 0$, $b = \max_j |v_j| = |v_k|$. Then

$$\rho(A)b = |\lambda||v_k| = \Big|\sum_j A_{kj}v_j\Big| \le \sum_j A_{kj}|v_j| \le \sum_j A_{kj}b = \rho_k(A)b \le \rho(A)b.$$

So equality holds throughout. Since all $A_{kj} > 0$, $|v_j| = b$ for all j, and the numbers $A_{kj}v_j$ all have the same argument (equality in the triangle inequality), so all v_j equal some $z \neq 0$: $v = zu$. Then $\lambda zu = Azu = z(\rho_1(A), \ldots, \rho_n(A))^t$, so every $\rho_i(A) = \lambda$, i.e. $\lambda = \rho(A)$. Every eigenvector for λ is a multiple of u. ■

Corollary 1

If A has positive entries and $|\lambda| = \nu(A)$ for an eigenvalue λ, then $\lambda = \nu(A)$ and $\dim E_\lambda = 1$.

Proof. Apply Theorem 5.18 to A^t ($\rho(A^t) = \nu(A)$), and note $\operatorname{rank}(A - \lambda I) = \operatorname{rank}(A^t - \lambda I)$. ■

Corollary 2

If A is a transition matrix with positive entries, every eigenvalue $\lambda \neq 1$ has $|\lambda| < 1$, and $\dim E_1 = 1$.

Proof. $\nu(A) = 1$. By Corollary 3 of Theorem 5.16, $|\lambda| \leq 1$, and if $|\lambda| = 1$ then $\lambda = 1$ by Corollary 1. Also $\dim E_1 = 1$ by Corollary 1. ■

Theorem 5.19

Let A be a regular transition matrix and λ an eigenvalue. Then (a) $|\lambda| \leq 1$;
(b) if $|\lambda| = 1$, then $\lambda = 1$ and $\dim E_\lambda = 1$.

Proof.

(a) Corollary 3 of Theorem 5.16.

(b) Let A^k be positive; then so is $A^{k+1} = A \cdot A^k$ (each row of A has a positive entry, since otherwise $A^k = A \cdot A^{k-1}$ would have a zero row). If $Av = \lambda v$, $|\lambda| = 1$, then λ^k and λ^{k+1} are eigenvalues of modulus 1 of positive transition matrices, so both equal 1 (Corollary 2 above), and $\lambda = \lambda^{k+1}/\lambda^k = 1$. Finally $E_1(A) \subseteq E_1(A^k)$, which has dimension 1. ■

Corollary

A diagonalizable regular transition matrix A has a limit $\lim A^m$.

Proof. Theorems 5.19 and 5.14. ■

Theorem 5.20

Let A be an $n \times n$ regular transition matrix. Then
(a) 1 has multiplicity 1 as an eigenvalue;
(b) $L = \lim A^m$ exists;
(c) L is a transition matrix;
(d) $AL = LA = L$;
(e) all columns of L equal the unique probability vector v with $Av = v$;
(f) $\lim A^m w = v$ for every probability vector w.

Proof.

(a) [Supplement] (The book defers this to Chapter 7.) Transition matrices have entries in $[0, 1]$, so $\{A^m w\}$ is bounded for each $w \in \mathbb{C}^n$. Suppose $(A - I)^2 w = 0$. Then $u' = (A - I)w \in E_1$ and $A^m w = w + mu'$, so boundedness forces $u' = 0$. Thus $\mathrm{N}((A - I)^2) = \mathrm{N}(A - I)$. Then $\mathrm{N}(A - I) \cap \mathrm{R}(A - I) = \{0\}$: if $x = (A - I)y$ and $(A - I)x = 0$, then $y \in \mathrm{N}(A - I)$ and $x = 0$. By

the dimension theorem $\mathbb{C}^n = \mathrm{N}(A-I) \oplus \mathrm{R}(A-I)$, both A-invariant, and 1 is not an eigenvalue of A on $\mathrm{R}(A-I)$. By Theorem 5.24 the multiplicity of 1 is $\dim \mathrm{N}(A-I) = 1$ (Theorem 5.19).

(b) By (a) and Theorem 5.19, the conditions of Theorem 5.13 hold.

(c) A^m is a transition matrix, so $u^t L = \lim u^t A^m = u^t$; entries stay nonnegative.

(d) $AL = \lim A^{m+1} = L$ by Theorem 5.12, and likewise $LA = L$.

(e) By (d) each column of L lies in E_1, and by (c) it is a probability vector. As $\dim E_1 = 1$, there is exactly one probability vector v in E_1.

(f) $y = \lim A^m w = Lw$ is a probability vector with $Ay = ALw = Lw = y$, so $y = v$ by (e). ■

v is the **fixed probability vector** (**stationary vector**) of A.

Example 4

Preferences (bread, wine, "thou") with initial distribution $P = (0.50, 0.30, 0.20)^t$ and transition matrix $A = \begin{pmatrix} 0.40 & 0.20 & 0.20 \\ 0.10 & 0.70 & 0.20 \\ 0.50 & 0.10 & 0.60 \end{pmatrix}$. A is positive, with eigenvalues $1, 0.5, 0.2$, and $AP = (0.30, 0.30, 0.40)^t$. The stationary vector is $v = (0.25, 0.35, 0.40)^t$: in the long run 25%, 35%, 40%, whatever the initial distribution.

Example 5

Crop rotation (corn, soybeans, wheat; half of each crop's land switches to each other crop): $A = \frac{1}{2}\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, $P = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})^t$ from $300, 200, 100$ acres. A^2 is positive, the eigenvalues are $1, -\frac{1}{2}, -\frac{1}{2}$, and the stationary vector is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})^t$. Eventually $600 \cdot \frac{1}{3} = 200$ acres go to each crop.

An **absorbing** state (one never left) leads to non-regular chains; the limit can still exist by Theorem 5.13.

5.4 Invariant subspaces and the Cayley–Hamilton theorem

Motivation. Even non-diagonalizable operators can be broken into pieces on invariant subspaces. Cyclic subspaces $\mathrm{span}\{v, \mathsf{T}v, \mathsf{T}^2 v, \dots\}$ have computable characteristic polynomials, which yields the Cayley–Hamilton theorem: every operator is annihilated by its characteristic polynomial.

Definition (Invariant subspace)

$W \subseteq V$ is **T-invariant** if $\mathsf{T}(W) \subseteq W$. Then $\mathsf{T}_W : W \to W$ denotes the restriction.

Example 1

$\{0\}$, V, $\mathrm{R}(\mathsf{T})$, $\mathrm{N}(\mathsf{T})$ and every eigenspace E_λ are T-invariant.

Example 2

For $\mathsf{T}(a,b,c) = (a+b, b+c, 0)$, the xy-plane $\{(x,y,0)\}$ and the x-axis $\{(x,0,0)\}$ are T-invariant.

Definition (Cyclic subspace)

For $x \in V$, the **T-cyclic subspace generated by** x is $W = \operatorname{span}\{x, \mathsf{T}(x), \mathsf{T}^2(x), \dots\}$, the smallest T-invariant subspace containing x.

Example 3

For $\mathsf{T}(a,b,c) = (-b+c, a+c, 3c)$: $\mathsf{T}(e_1) = e_2$, $\mathsf{T}(e_2) = -e_1$, so the cyclic subspace generated by e_1 is $\operatorname{span}\{e_1, e_2\}$.

Example 4

$\mathsf{T}(f) = f'$ on $\mathsf{P}(\mathbb{R})$: the cyclic subspace generated by x^2 is $\operatorname{span}\{x^2, 2x, 2\} = \mathsf{P}_2(\mathbb{R})$.

Theorem 5.21

If W is T-invariant (V finite-dimensional), the characteristic polynomial of T_W divides that of T.

Proof. Extend a basis γ of W (k vectors) to a basis β of V. Then $[\mathsf{T}]_\beta = \begin{pmatrix} B_1 & B_2 \\ O & B_3 \end{pmatrix}$ with $B_1 = [\mathsf{T}_W]_\gamma$, and $f(t) = \det(B_1 - tI)\det(B_3 - tI)$ (Section 4.4, fact 8). ■

Example 5

$\mathsf{T}(a,b,c,d) = (a+b+2c-d,\ b+d,\ 2c-d,\ c+d)$ and $W = \operatorname{span}\{e_1, e_2\}$, which is invariant. $[\mathsf{T}_W] = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has characteristic polynomial $(t-1)^2$, which divides $f(t) = (t-1)^2(t^2-3t+3)$.

Theorem 5.22

Let W be the T-cyclic subspace generated by $v \neq 0$ and $k = \dim W$ (V finite-dimensional). Then
(a) $\{v, \mathsf{T}(v), \dots, \mathsf{T}^{k-1}(v)\}$ is a basis of W;
(b) if $a_0 v + a_1 \mathsf{T}(v) + \cdots + a_{k-1}\mathsf{T}^{k-1}(v) + \mathsf{T}^k(v) = 0$, the characteristic polynomial of T_W is $(-1)^k(a_0 + a_1 t + \cdots + a_{k-1}t^{k-1} + t^k)$.

Proof.

(a) Let j be the largest integer with $\beta = \{v, \dots, \mathsf{T}^{j-1}(v)\}$ independent ($j \geq 1$ as $v \neq 0$). Then $\mathsf{T}^j(v) \in Z = \operatorname{span}(\beta)$ by Theorem 1.7. Z is T-invariant: T maps $\mathsf{T}^i(v)$ to $\mathsf{T}^{i+1}(v) \in Z$ for $i < j$. So $Z \supseteq W$ (the smallest invariant subspace containing v), $Z = W$, and $j = k$.

(b) In the basis β of (a), $[\mathsf{T}_W]_\beta$ has $e_2, \dots, e_k$ as its first $k-1$ columns and $(-a_0, \dots, -a_{k-1})^t$ as its last column. Expanding $\det([\mathsf{T}_W]_\beta - tI)$ along the last column (or by induction on k) gives $(-1)^k(a_0 + a_1 t + \cdots + t^k)$. ■

Example 6

For T of Example 3 and $W = \operatorname{span}\{e_1, e_2\}$: $\mathsf{T}^2(e_1) = -e_1$, so $e_1 + \mathsf{T}^2(e_1) = 0$ and the characteristic polynomial of T_W is $t^2 + 1$.

Theorem 5.23 (Cayley–Hamilton)

If $f(t)$ is the characteristic polynomial of T on finite-dimensional V, then $f(\mathsf{T}) = \mathsf{T}_0$.

Proof. We show $f(\mathsf{T})(v) = 0$ for every $v \neq 0$. Let W be the T-cyclic subspace generated by v, $k = \dim W$, and a_i as in Theorem 5.22(b). Then $g(t) = (-1)^k(a_0 + \cdots + t^k)$ is the characteristic polynomial of T_W, and $g(\mathsf{T})(v) = (-1)^k(a_0 v + \cdots + \mathsf{T}^k(v)) = 0$. By Theorem 5.21, $f = qg$, so $f(\mathsf{T})(v) = q(\mathsf{T})(g(\mathsf{T})(v)) = 0$. ■

Example 7

$\mathsf{T}(a,b) = (a+2b, -2a+b)$, $A = \left(\begin{smallmatrix} 1 & 2 \\ -2 & 1 \end{smallmatrix}\right)$, $f(t) = t^2 - 2t + 5$: $A^2 - 2A + 5I = \left(\begin{smallmatrix} -3 & 4 \\ -4 & -3 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 2 & 4 \\ -4 & 2 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 5 & 0 \\ 0 & 5 \end{smallmatrix}\right) = O$.

Corollary (Cayley–Hamilton for matrices)

If f is the characteristic polynomial of $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, then $f(A) = O$.

Proof. $f(\mathsf{L}_A) = \mathsf{L}_{f(A)}$ is the zero map by Theorem 5.23; apply Theorem 2.15(b). ■

5.4.1 Invariant subspaces and direct sums

Theorem 5.24

If $V = W_1 \oplus \cdots \oplus W_k$ with each W_i T-invariant and f_i the characteristic polynomial of T_{W_i}, then $f_1 \cdots f_k$ is the characteristic polynomial of T.

Proof. Induction on k; $k = 1$ is trivial. For $k = 2$, take bases β_1, β_2 of W_1, W_2; $\beta = \beta_1 \cup \beta_2$ is a basis (Theorem 5.10) and $[\mathsf{T}]_\beta = \left(\begin{smallmatrix} [\mathsf{T}_{W_1}] & O \\ O & [\mathsf{T}_{W_2}] \end{smallmatrix}\right)$, whose characteristic polynomial is $f_1 f_2$ (Section 4.4, fact 8). For $k > 2$, $W = W_1 \oplus \cdots \oplus W_{k-1}$ is T-invariant with $V = W \oplus W_k$; apply the case $k = 2$ and induction to T_W. ■

Example 8

$\mathsf{T}(a,b,c,d) = (2a-b, a+b, c-d, c+d)$, $W_1 = \mathrm{span}\{e_1, e_2\}$, $W_2 = \mathrm{span}\{e_3, e_4\}$: $f_1 = t^2 - 3t + 3$, $f_2 = t^2 - 2t + 2$, and $f = f_1 f_2$.

Definition (Direct sum of matrices)

For $B_1 \in \mathrm{M}_{m\times m}(\mathbb{F})$, $B_2 \in \mathrm{M}_{n\times n}(\mathbb{F})$, $B_1 \oplus B_2 = \left(\begin{smallmatrix} B_1 & O \\ O & B_2 \end{smallmatrix}\right)$; inductively $B_1 \oplus \cdots \oplus B_k = (B_1 \oplus \cdots \oplus B_{k-1}) \oplus B_k$, block diagonal.

Example 9

$\left(\begin{smallmatrix} 1 & 2 \\ 1 & 1 \end{smallmatrix}\right) \oplus (3) \oplus \left(\begin{smallmatrix} 1 & 2 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{smallmatrix}\right)$ is the 6×6 block-diagonal matrix with these blocks on the diagonal.

Theorem 5.25

If $V = W_1 \oplus \cdots \oplus W_k$ with T-invariant W_i, ordered bases β_i of W_i, $\beta = \beta_1 \cup \cdots \cup \beta_k$, $A = [\mathsf{T}]_\beta$ and $B_i = [\mathsf{T}_{W_i}]_{\beta_i}$, then $A = B_1 \oplus \cdots \oplus B_k$.

Proof. Induction on k. For $k = 2$: if $v \in \beta_1$, then $\mathsf{T}(v) \in W_1$ is a combination of β_1 alone, with coordinates the corresponding column of B_1; similarly for β_2. So $A = B_1 \oplus B_2$. For $k > 2$ apply this to $W = W_1 \oplus \cdots \oplus W_{k-1}$ and W_k, and use induction on T_W. ■